preparations, challenges & roadmap in India's endeavour to vaccinate its citizens.'[8]

On the morning of 28 November, the prime minister boarded an Indian Air Force (IAF) flight from New Delhi to Ahmedabad followed by a quick chopper ride to his first destination at Zydus Biotech Park in Changodar, Gujarat, to review Zydus Cadila's candidate ZyCoV-D. He was warmly welcomed by the Patel family following which Chairman Pankaj Patel, his son and joint managing director at Zydus Cadila, Sharvil Patel and their chief scientist briefed him on the technology platform on which their vaccine was based. If approved, it would be the first DNA plasma vaccine in the world, making a momentous contribution to the scientific community. They also briefed him about how the vaccine which was intradermal would be administered via a painless patch versus a needle syringe. They demonstrated a wide gamut of pharmaceutical products such as anti-viral drugs including remdesivir which they were working on and offering to combat COVID-19. After the briefing, they escorted the PM on a tour of the production unit and demonstrated the nitty-gritties of the manufacturing process for the vaccine. Since it was a packed trip, the PM immediately boarded the chopper back to Ahmedabad airport from where he boarded the IAF plane, had a quick in-flight lunch and took a short car ride after deplaning at Hakimpet IAF airbase, to the Bharat Biotech facility in Genome Valley in the Shameerpet area of Telangana's capital city Hyderabad.[9]

Reaching the Bharat Biotech facility at about two in the afternoon, he was welcomed by Chairman and Managing Director Krishna Ella, his wife and Joint Managing Director Suchitra Ella and their chief scientist. The team gave him an overview of the company including details on how they had been making rotavirus vaccines and supplying them to low-income countries. They broke their briefing down in four parts.

In the first part, they discussed the roll-out of their current vaccine candidate, Covaxin, jointly developed by ICMR, which was based on an inactivated virus. The second part that they discussed was the development, trial and approval of their nasal vaccine. They were very hopeful about this vaccine as it would provide sterilizing immunity. To understand what that meant, let's bring in some scientific context here. The COVID-19 virus is known to heavily target cells in the lungs. When a standard vaccine is administered, it first enters the lungs after which, depending on the